

Carl Bryant Carlos

BS Computer Science (AI), Mapúa University

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Residency: National Capital Region, Philippines



CERTIFICATIONS AND AREAS OF INTEREST

- ❖ AWS Certified Solutions Architect – Associate
- ❖ AWS Certified AI Practitioner
- ❖ AWS Certified Cloud Practitioner
- ❖ TOEIC Score: 945 (CEFR C1)

Main interests: Networks, Cloud Computing, IT, DevOps, Artificial Intelligence, Cybersecurity

TECHNICAL SKILLS

Programming & Development

- Python, Java, C/C++, SQL, Bash, HTML/CSS
- **Machine Learning & AI:** Scikit-learn, Orange, Jupyter

Cloud & DevOps

- **AWS:** VPC, subnets, EC2, S3, IAM, security groups, route tables, Lambda, RDS – architecting secure, scalable, and fault-tolerant distributed systems (SAA certified)
- **CI/CD:** Docker, GitHub Actions, Node-RED
- **Virtualization & Servers:** Nginx, Apache, VPS Management (Linux-based), VirtualBox, WSL
- **IoT & Edge Computing:** Raspberry Pi 5 (ARM64) and Arduino (Embedded Systems)

Networking & Security

- **Protocols & Concepts:** OSI and TCP/IP, CGNAT, NAT, DNS, DHCP, MQTT, SNMP, LoRaWAN, HTTP/HTTPS, firewall, subnetting, tunneling, proxies
- **Tools:** Wireshark, Nmap/Zenmap, Netmonster, Cisco Packet Tracer, PRTG
- **VPN & Remote Access:** Wireguard, OpenVPN, MobaXterm, PuTTY, Termux, RealVNC

Software Tools & Platforms

- **Development:** Visual Studio Code, Git/GitHub, Postman
 - **Productivity:** MS Office, Canva, Figma, AnyDesk, TeamViewer, Parsec
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WORK EXPERIENCE

IT Intern (R&D)

(Nov 2025 – Jan 2026 @ Geoplan Philippines, Inc.)

- Evaluated and configured IoT devices to assess product viability for PoC proposals and client deployments.
 - Assisted in on-site client PoC, integrated IoT devices into their PRTG monitoring platform via Node-RED workflows, and led troubleshooting of network connectivity issues.
 - Deployed a containerized IoT dashboard using Docker, integrating real-time device data ingestion via webhooks from LoRaWAN sensors.
 - Diagnosed a UWB Real-Time Location System's inconsistent connectivity and narrowed the root cause to protocol and cloud latency, prompting a server switch to on-premises testing.
 - Created technical documentation and tutorials in Google Slides to support internal knowledge base and client-facing presentations.
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RESEARCH PAPERS AND PROJECTS HANDLED

Binary Occupancy Detection: Utilizing Wi-BFI and Few-Shot Learning via Consumer Devices

(Ongoing Thesis, 2026 @ Mapúa University)

- Detects room occupancy using only Wi-Fi signals; no cameras needed
- Few-shot learning for domain-agnostic meta-learning; faster space calibrations
- Combines Machine Learning (AI) with Wi-Fi 5 and above (BFM/BFI)

TDA Classification using SVM for the Philippine Stock Exchange Index

(Unpublished Research Activity for AI-2 Course, 2024 @ Mapúa University)

- Classifies the market trend of the PSEI using topological data analysis and machine learning
- Pipeline is created using the giotto-tda Python library: Taken's embedding, Vietoris-Rips persistence homology, amplitude, and support vector machine with a radial basis function kernel; hyperparameter tuning using GridSearchCV was discontinued
- The idea is "Point clouds tend to form clusters during market crashes."

TReE: Triple Renewable Energy (Solar, Water, and Wind) as a Power Source of Weather Monitoring System

(Research Project, 2022 @ Pasig City Science Highschool)

- Handled the Arduino programming for the DHT22 sensor's readings display
- Combines the three renewable energies into one generator

TRAININGS

Full list (30+) at LinkedIn: <https://www.linkedin.com/in/carlcarlos>

CCNA: Enterprise Networking, Security, and Automation, Cisco	– 08/25
Foundations of Cybersecurity, Google	– 03/25
Introduction to Cloud Computing, IBM	– 02/25
Introduction to DevOps, IBM	– 02/25
BI Foundations with SQL, ETL and Data Warehousing, IBM	– 05/24

EDUCATION

Mapúa University - Makati Campus

Bachelor of Science in Computer Science with Specialization in Artificial Intelligence
(2022 – Ongoing)
Expected Graduation: August, 2026 (Remaining course: Thesis)
GWA / Running Average: 1.7820

Pasig City Science Highschool

Science, Technology, Engineering, and Mathematics (2020 – 2021)

PERSONAL DATA

- Credly: <https://www.credly.com/users/carl-bryant-carlos>
- Full name: Carl Bryant Astacaan Carlos
- Nationality: Filipino